F22KN411L-XE

R W B N

Q102

F102

TOROID NOTE 4

L1 T1 / L2 T2 / L3 T3 / PE

F22KN411-JB24

F22-KN-411L

CIL4 CARBON REGEN KILN HEATERS
ZONE 2 BANK 4
168 kW

630A

LINE 125

250A
250A

LINE 112

A B

C70 C71

LINE 142

THYRO-PX
3PX-500-280 HF

F106 2A 32A

F108 6A 32A

1061 1062 1063

N PE L1 L2 L3

LLI RD

Q110 2A 1100

Q114 2A 1160

K114 240VN

FROM LINE 105

X7
240VAC SUPPLY

DI: 03/27
DIGITAL INPUT
SLOT No. 03
1769-IQ32

F22KN411-CP
(LOCATED IN THE FIELD)

1162 X1 25 K114 26 1164

11 X1 28 27

12

K1 ALARM 1 2 9 1167

24VDC CONTROL

TO NEXT MODULE

Q120 4A

1200

LINE 146

COMMISSIONING BYPASS

K114

1230

INTERNAL WIRING TO F22KN411-CP
FIELD CABLE MARSHALLING TERMINALS
F22-SW-031 TIER 01F
2000-F22-TER-EL-10253

16 15

21 20

X51
1 +24VDC
2 +24VDC 1194
3 IN 1.1 1204
4 IN 1.2
5 IN 1.3
6 IN 1.4
7 IN 1.5
8 IN 1.6
9 COMMON

24VDC AUXILIARY SUPPLY -X3

0VDC

X52
1 +5VDC
2 AI 1.1
3 AI 1.2
4 AI 1.3
5 COMMON
6 AO 1.1
7 AO 1.2
8 AO 1.3
9 COMMON

CONT'D ON 2000-F22-SCM-EL-10261

ASI451141
ANALOGUE SIGNAL SPLITTER
1 INPUT/ 4 OUTPUT

X1 41 1334

X1 42 1344

OUT 2 + 24 - 22

CONT'D ON 2000-F22-SCM-EL-10265
SIGNAL SPLITTER
(LOCATED IN TIER IF CABLE ZONE)

TO F22KN411J-XE
2000-F22-SCM-EL-10261

TO F22KN411N-XE
2000-F22-SCM-EL-10265

CAT6 — CAT6 — CAT6

@ASM MODULE

-X1201 ANYBUS XX

CAT6 — CAT6 — CAT6

TO SWITCHROOM COMMS RACK
MODBUS ETHERNET SWITCH
F22-ETH-001
2000-F22-TER-IC-10003

(MOUNTED ON SWITCHROOM WALL)

LINE 106

(DOOR MOUNTED)

K142

TEST C70

RESET C71

TRIP 18 17 20 21

NHP RD1E21HBN (IME DELTA 72-S)
NOTE 5

Q102 1453

LINE 121

MCCB SHUNT TRIP COIL 102

EARTH LEAKAGE RELAY

DRIVE ISOLATED RELAY 116, 123

HEATER RUNNING

ZONE 2 BURST FIRE ENABLE

SPARE INPUT (FAULT ACKNOWLEDGE)

SPARE INPUT

SPARE INPUT

SPARE INPUT

SPARE INPUT

SPARE ANALOGUE INPUT

SPARE ANALOGUE INPUT

ZONE 2 FIRING ANALOGUE INPUT

SPARE ANALOGUE OUTPUT

SPARE ANALOGUE OUTPUT

SPARE ANALOGUE OUTPUT

PANEL TERMINAL BLOCK

TO LINE 147 — F22KN411LXE-P3 2.5 mm² 2C+E	RD	1	F22KN411L-1110
	BK	2	F22KN411L-240VN
	GNYE	3	EARTH
TO LINE 128 — F22KN411LXE-C1 1.5 mm² 10C+E		4	F22KN411L-1230
		5	F22KN411L-0VDC
		6	F22KN411L-1194
		7	F22KN411L-1204
		8	F22KN411L-1164
	GNYE	9	F22KN411L-1167
TO F22KN411-CP FIELD CABLE MARSHALLING TERMINALS F22-SW-031 TIER 01F 2000-F22-TER-EL-10253 — F22KN411LXE-I1 1.5 mm² 2 Pair O/S	1WH	10	EARTH
	1BK	11	F22KN411L-1334
		12	F22KN411L-1344
		13	
		14	SPARE
		15	SPARE
		16	SPARE
		17	SPARE
	2WH	18	SPARE
	2BK	19	SPARE
		20	SCREEN

F22KN411L-XE DEVICE TERMINALS

-X7 230VAC POWER

L1	F22KN411L-1100
N	F22KN411L-240VN

-X3 AUX POWER

+24VDC	F22KN411L-1230
-24VDC	F22KN411L-0VDC

-X51 I/O CARD

+24VDC	
+24VDC	F22KN411L-1194
IN 1.1	F22KN411L-1204
IN 1.2	
IN 1.3	
IN 1.4	
IN 1.5	
IN 1.6	
COMMON	

-X52 I/O CARD

+5VDC	
AI 1.1	
AI 1.2	
AI 1.3	F22KN411L-1334
COMMON	F22KN411L-1344
AO 1.1	
AO 1.2	
AO 1.3	
+5VDC	

K1 FAULT RELAY

1	F22KN411L-1164
2	F22KN411L-1167
3	

CABLE ZONE 24VDC TERMINALS

24VDC LOOPED POWER NEXT MODULE / PREV MODULE

Q120 4A		F22SW031-2280
1		F22SW031-2290
2		F22SW031-2290
3		F22KN411L-1200
4		F22KN411L-1230
5		F22KN411L-1230
6		F22KN411L-0VDC
7		
8		
9		
10		
11		
12		
13		
14		
15		F22KN411L-1194
16		F22KN411L-1194
17		
18		
19		
20		F22KN411L-1204
21		F22KN411L-1204
22		
23		
24		
25		F22KN411L-1162
26		F22KN411L-1164
27		F22KN411L-1167
28		F22KN411L-1167
29		EARTH
30		EARTH

COMMISSIONING BYPASS

NOTE 2

F22KN411LXE-C1 1.5 mm² 10C+E — TO LINE 110

INTERNAL WIRING TO F22KN411-CP
FIELD CABLE MARSHALLING TERMINALS
F22-SW-031 TIER 01F
2000-F22-TER-EL-10253

240VAC TERMINALS

1		F22KN411L-1100
2		F22KN411L-240VN
3		
4		EARTH

NOTES 2, 6

F22KN411LXE-P3 2.5 mm² 2C+E — TO LINE 108

NOTES

1. TERMINALS ON TERMINAL STRIPS ABOVE 50VAC SHALL HAVE A SEPARATE PERSPEX SHROUD AND SHALL BE LABELLED "DANGER xxxVAC".
2. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F22KN411L-1200, UNLESS OTHERWISE NOTED. (1)
3. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
4. NHP TDGE2 IN:250A IΔN MIN:0.3A TOROID TO SUIT CABLE DIAMETER.
5. EARTH LEAKAGE RELAY (ELV) USER INTERFACE TO BE MOUNTED ON MCC MODULE DOOR. RELAY ENERGISED (NORMALLY CLOSED CONTACT OPEN) IN NORMAL OPERATION. (1)
6. 240VAC TERMINALS SHALL BE LOCATED IN THE MCC DRIVE MODULE.
7. F22KN411-CP FIELD CABLE MARSHALLING TERMINAL BLOCK IS DENOTED BY MARSHALLING TERMINALS LABELLED X1 (1)

LEGEND

- 00/01/00 — DCS DIGITAL INPUT
- 00/02/00 — DCS DIGITAL OUTPUT
- MCC TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- (x) DEVICE TERMINAL
- (cc) INDICATION LAMP cc = COLOUR

(1)

TERMINATION LEGEND

Q120 4A			F22SW031-2xxx
1			F22SW031-2xxx
2			F22SW031-2xxx
3			F22KN411L-1200
4			F22KN411L-1230
5			F22KN411L-1230
6			EARTH

RD / BK / GNYE

- CORE COLOUR/ IDENTIFIER
- BARRIER
- FERRULE
- JUMPER/DISCONNECT/ FUSE/EARTH
- TERMINAL No.
- DESTINATION

2000-F22-SCM-EL-10263_Rev1

F22-SW-031 - F22-KN-411L carbon regen kiln htrs
zone 2 bank 4 - schematic and termination diagram